# PolicyGraphAudit-RT

Compliance Audit Report
Heterogeneous GNN Privacy Analysis

# Amsterdam Offline City Map

`com.ulmon.android.playamsterdamofflinemap`

| | |
|---|---|
| **Developer** | CityMaps2Go |
| **Genre** | TRAVEL_AND_LOCAL |
| **Installs** | 100,000+ |
| **Audit Date** | 2026-04-26 |

| Overall Risk Score | 0.42 | MEDIUM |
|---|---|---|

| 5 | 1 | 4 | 4 |
|---|---|---|---|
| Consistent | Mismatches | Over-disclosure | Undeclared |

Research prototype · Ongoing research | Policy: http://www.ulmon.com/support/privacy-policy/

## Executive Summary

**Amsterdam Offline City Map** declares 5 data type(s) collected and 3 shared in its Play Data Safety label, across 14 data type(s) analyzed. Our heterogeneous GNN audit model identifies **4 potential undeclared collection(s)** — data types implied by embedded SDK signals but absent from both label and policy — and **1 policy-vs-label mismatch(es)**. Overall risk score: **0.42 (MEDIUM)**.

**Top discrepancies by risk:**

- **installed_apps**: Undeclared Collection (confidence 0.98)
- **advertising_id**: Undeclared Collection (confidence 0.92)
- **crash_logs**: Undeclared Collection (confidence 0.82)

**Class definitions:**

**CONSISTENT**: Label, policy, and SDK signals agree.

**POLICY_LABEL_MISMATCH**: Label discloses data not mentioned in policy.

**OVER_DISCLOSURE**: Policy mentions data not declared in label.

**UNDECLARED_COLLECTION**: SDK signals collection undisclosed in label and policy.

## Discrepancy Table

Sorted by risk class (undeclared collection first). Y/N: label=data-safety label, policy=policy text, sdk=SDK signal.

| Data Type | Class | Conf. | Evidence |
|---|---|---|---|
| installed_apps | **Undeclared Collection** | 0.98 | label:N policy:N sdk:Y (1, 2, 3 +3) |
| advertising_id | **Undeclared Collection** | 0.92 | label:N policy:N sdk:Y (1, 2, 3 +3) |
| crash_logs | **Undeclared Collection** | 0.82 | label:N policy:N sdk:Y (1, 2, 3 +3) |
| diagnostics | **Undeclared Collection** | 0.79 | label:N policy:N sdk:Y (1, 2, 3 +3) |
| other_user_generated_content | **Policy Label Mismatch** | 0.98 | label:Y policy:N sdk:N (1, 2, 3 +3) |
| phone_number | **Over Disclosure** | 0.85 | label:N policy:Y sdk:N (1, 2, 3 +3) |
| precise_location | **Over Disclosure** | 0.84 | label:N policy:Y sdk:N (1, 2, 3 +3) |
| fitness_info | **Over Disclosure** | 0.73 | label:N policy:Y sdk:N (1, 2, 3 +3) |
| purchase_history | **Over Disclosure** | 0.56 | label:N policy:Y sdk:N (1, 2, 3 +3) |
| email_address | **Consistent** | 0.99 | label:Y policy:Y sdk:Y (1, 2, 3 +3) |
| app_interactions | **Consistent** | 0.99 | label:N policy:Y sdk:N (1, 2, 3 +3) |
| device_id | **Consistent** | 0.96 | label:Y policy:Y sdk:Y (1, 2, 3 +3) |
| name | **Consistent** | 0.94 | label:Y policy:Y sdk:Y (1, 2, 3 +3) |
| photos | **Consistent** | 0.92 | label:Y policy:Y sdk:N (1, 2, 3 +3) |

## Evidence Trails

Detailed evidence for UNDECLARED_COLLECTION or confidence > 0.85 non-CONSISTENT findings.

### Discrepancy: installed_apps — Undeclared Collection (confidence 0.98)

Label declares collection: **No**

Label declares sharing: **No**

Policy text mentions: **No**

SDK collection signals: **Yes** — 1, 2, 3, 5, 7 (+1 more)

Recommended action: Add 'installed_apps' to the Play Data Safety label under 'Data Collected' and update the privacy policy to disclose this collection.

### Discrepancy: advertising_id — Undeclared Collection (confidence 0.92)

Label declares collection: **No**

Label declares sharing: **No**

Policy text mentions: **No**

SDK collection signals: **Yes** — 1, 2, 3, 5, 7 (+1 more)

Recommended action: Add 'advertising_id' to the Play Data Safety label under 'Data Collected' and update the privacy policy to disclose this collection.

### Discrepancy: crash_logs — Undeclared Collection (confidence 0.82)

Label declares collection: **No**

Label declares sharing: **Yes**

Policy text mentions: **No**

SDK collection signals: **Yes** — 1, 2, 3, 5, 7 (+1 more)

Recommended action: Add 'crash_logs' to the Play Data Safety label under 'Data Collected' and update the privacy policy to disclose this collection.

### Discrepancy: diagnostics — Undeclared Collection (confidence 0.79)

Label declares collection: **No**

Label declares sharing: **Yes**

Policy text mentions: **No**

SDK collection signals: **Yes** — 1, 2, 3, 5, 7 (+1 more)

Recommended action: Add 'diagnostics' to the Play Data Safety label under 'Data Collected' and update the privacy policy to disclose this collection.

### Discrepancy: other_user_generated_content — Policy Label Mismatch (confidence 0.98)

Label declares collection: **Yes**

Label declares sharing: **No**

Policy text mentions: **No**

SDK collection signals: **No** — 1, 2, 3, 5, 7 (+1 more)

Recommended action: Update the privacy policy to disclose 'other_user_generated_content' collection, aligning it with the Play Data Safety label.

## Methodology

PolicyGraphAudit-RT represents each Android app as a tri-partite heterogeneous graph connecting three layers: **Privacy Policy** text (OPP-115 segment classification), **Play Data Safety Labels** (declared data types and purposes), and **Runtime Evidence** (inferred SDK trackers via Exodus Privacy). A 2-layer HeteroConv R-GCN encodes all three layers jointly; an MLP classifier head predicts discrepancy classes for each (App, DataType) pair.

The model was trained under an edge-masking protocol (30% of label-determining edges removed) to prevent circularity. It achieves **macro F1 = 0.956** on 39 held-out apps (UNDECLARED_COLLECTION F1 = 0.974), trained on 252 apps and 3,202 labeled pairs. Model card: https://github.com/PolicyGraphAudit-RT/reports/blob/main/reports/m5_model_card.md.

Known limitations: (1) Privacy labels are from a 2022 Play Store snapshot. (2) SDK presence is inferred via category-level priors, not measured from APK bytecode. (3) Policy classifier uses MiniLM + logistic regression (OPP-115 macro F1 = 0.70). (4) Labels are rule-derived weak supervision, not human-annotated. (5) English-language Android apps only.

## Disclaimer